

DTU



Perception for Autonomous Systems - 34759:

Image Feature Detection and Description

Lecturer: Evangelos Boukas—PhD

Image Features

- Feature Detection:
Find the most “prominent” Points (areas) in an image.
The ones which are likely to be detected in other images, as well
- Feature Description:
Create a “unique” descriptor fingerprint for each Feature point
- Feature Matching:
Find correspondences among different images

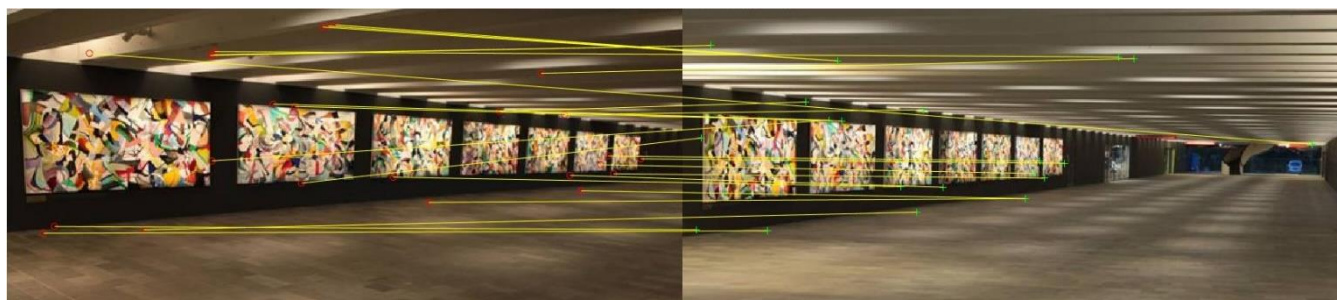
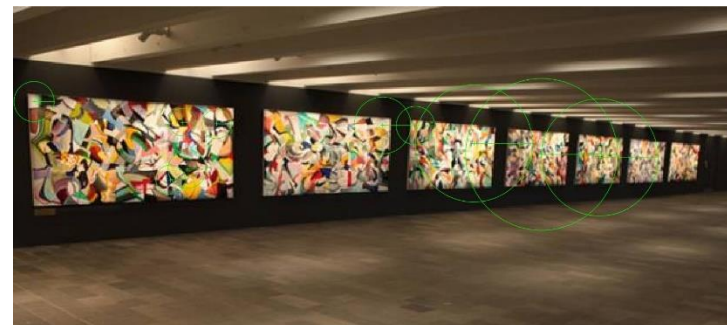
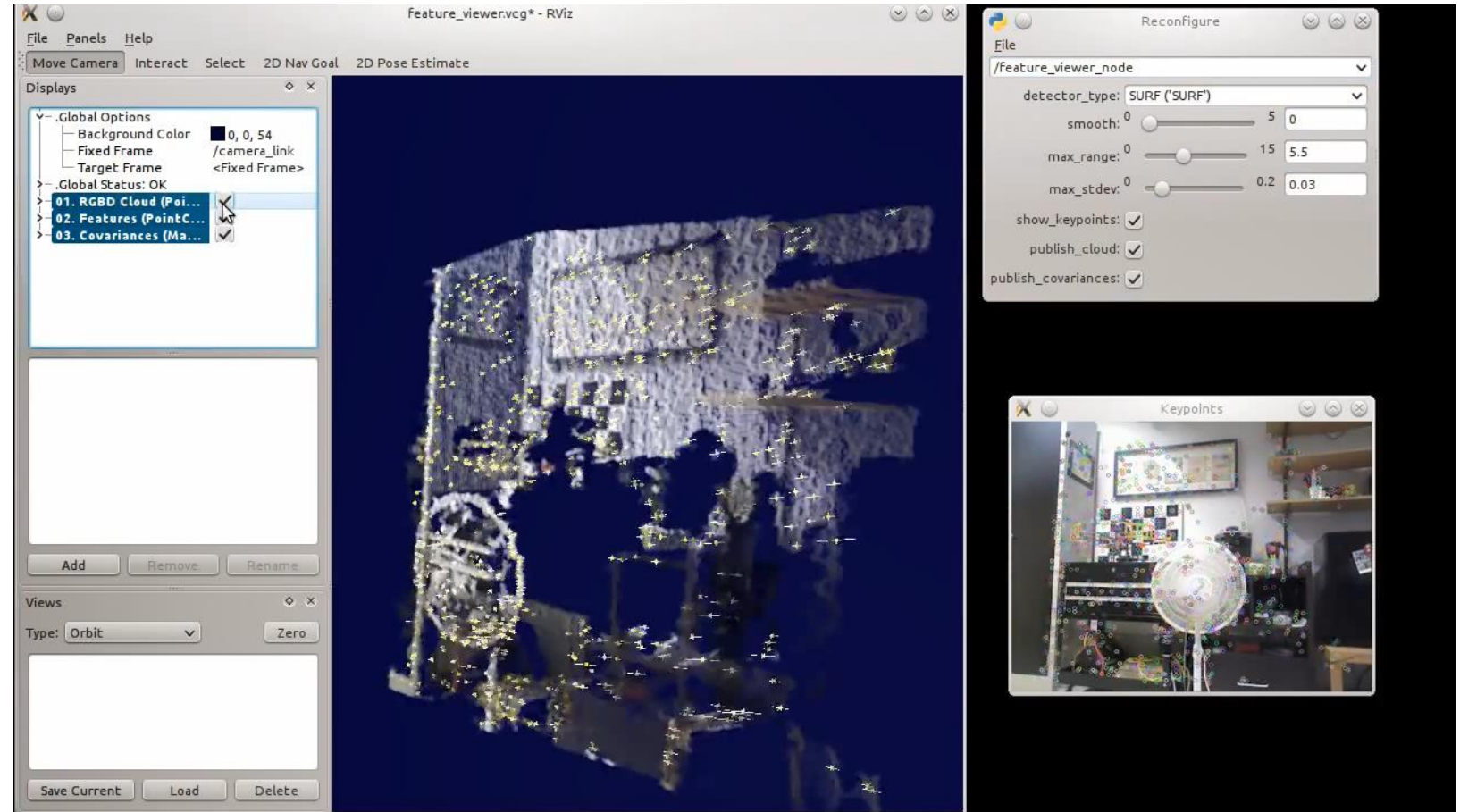


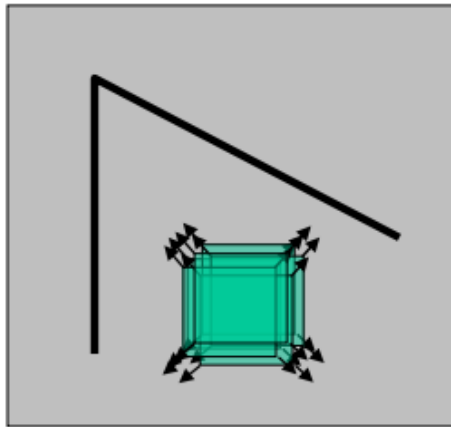
Image Features

- A quick view

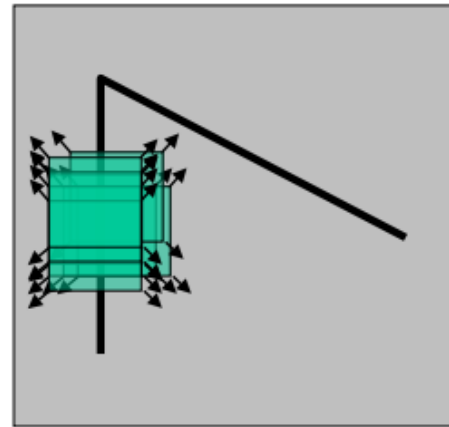


Feature Detection: Harris corner detector

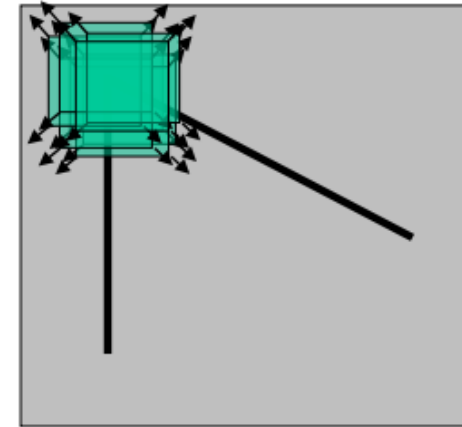
- Corners are great for features



“flat” region:
no change in
all directions



“edge”:
no change along
the edge direction

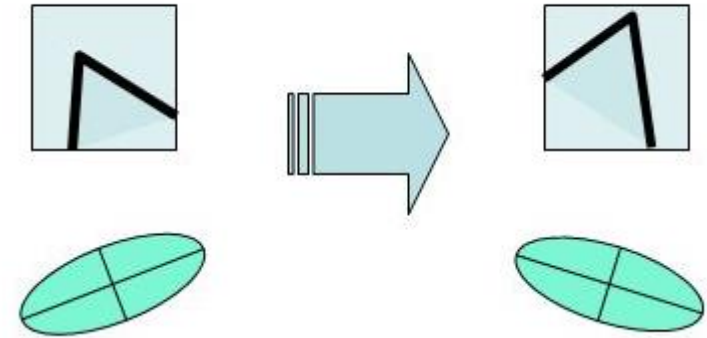


“corner”:
significant change
in all directions

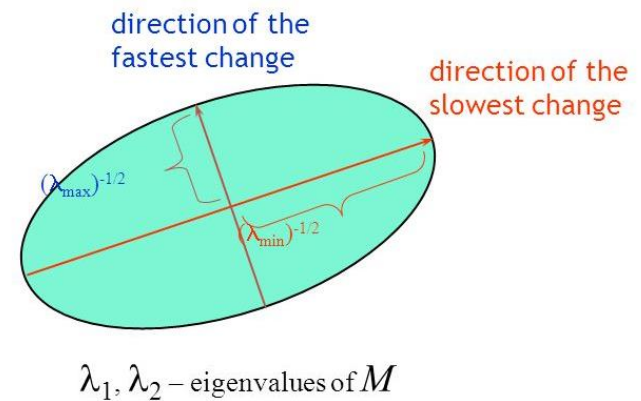
- Lines not so, why

Feature Detection: Harris corner detector

- Harris can discriminate among **edge**, **flat** and **corners**

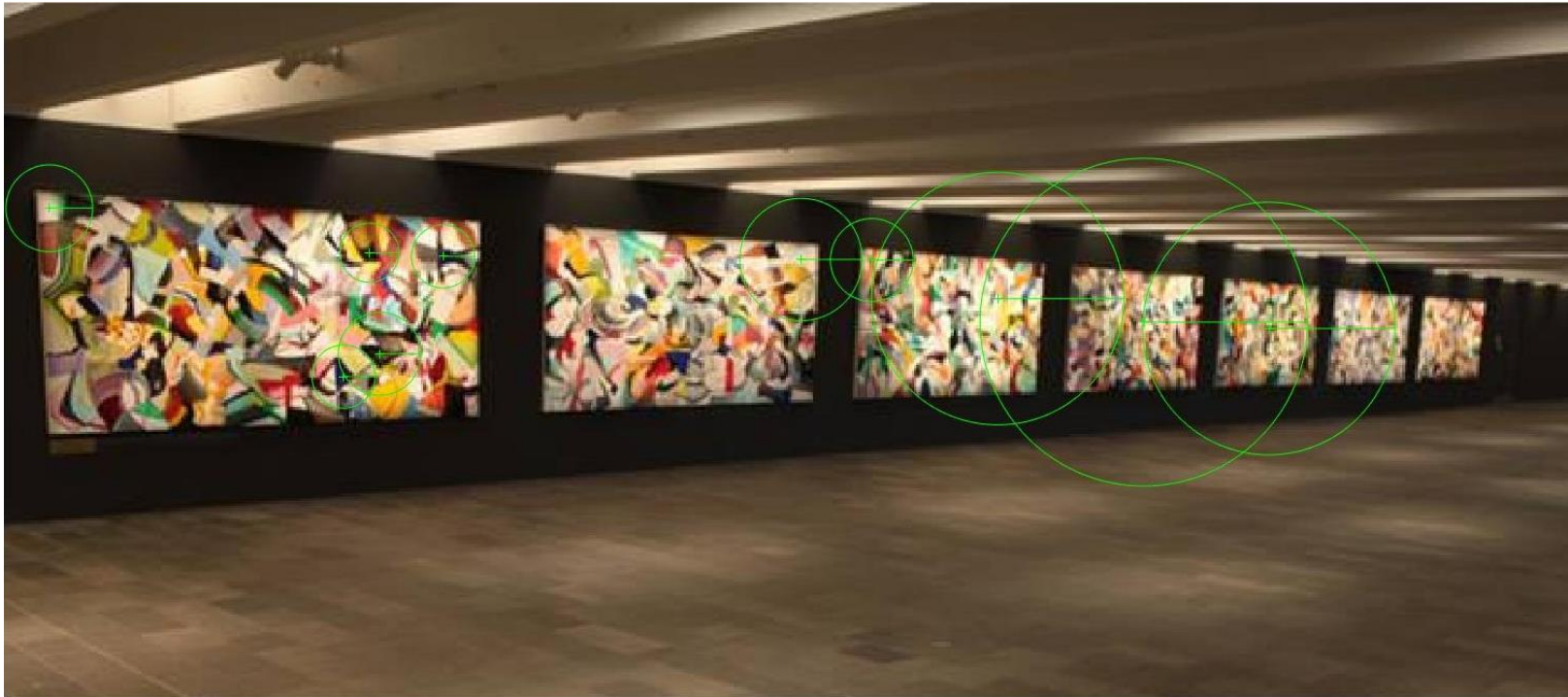


- Notice how by rotating it does not change
- Linear algebra



Feature Detection: Harris corner detector

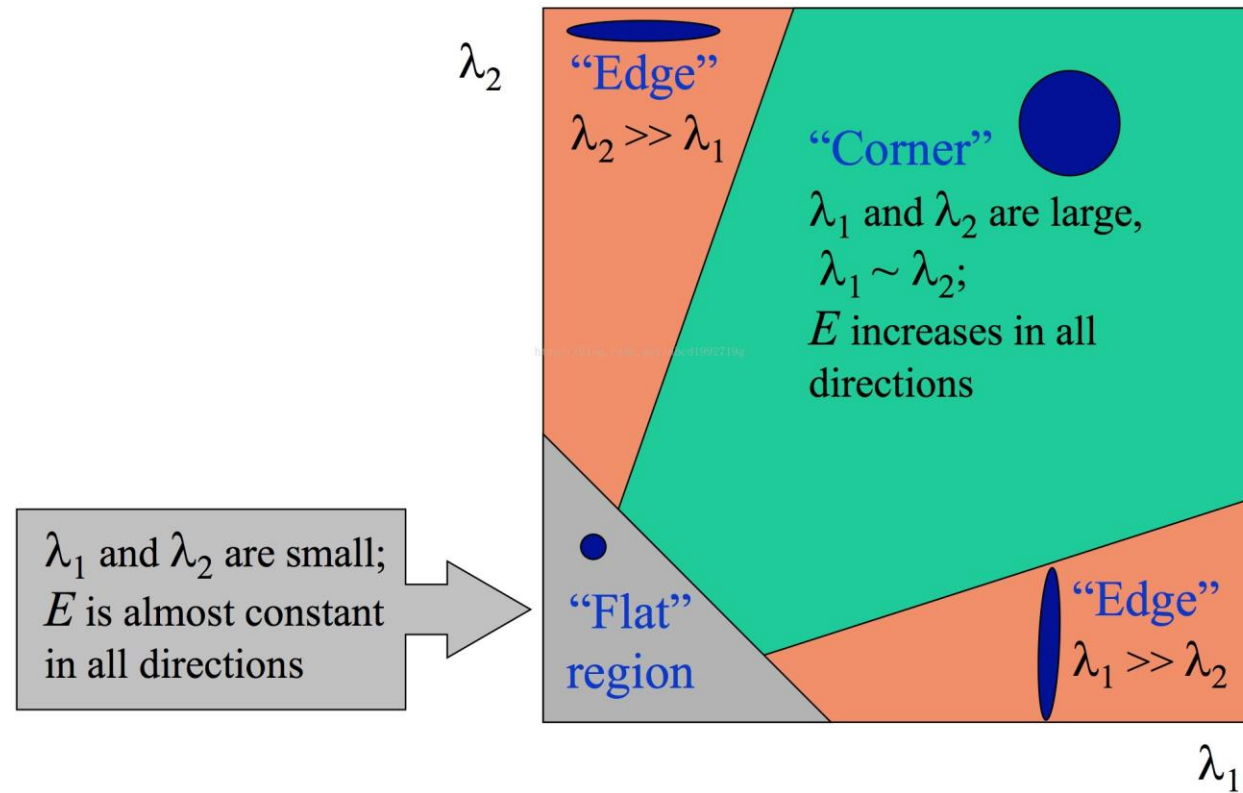
- So how are these eigenvalues useful?

 λ_1

Mikolajczyk, K., and Schmid, C., "A performance evaluation of local descriptors",
IEEE Transactions on Pattern Analysis and Machine Intelligence, 10, 27, pp 1615--1630, 2005.

Feature Detection: Harris corner detector

- So how are these eigenvalues useful?



Feature Detection: Scale Space Theory

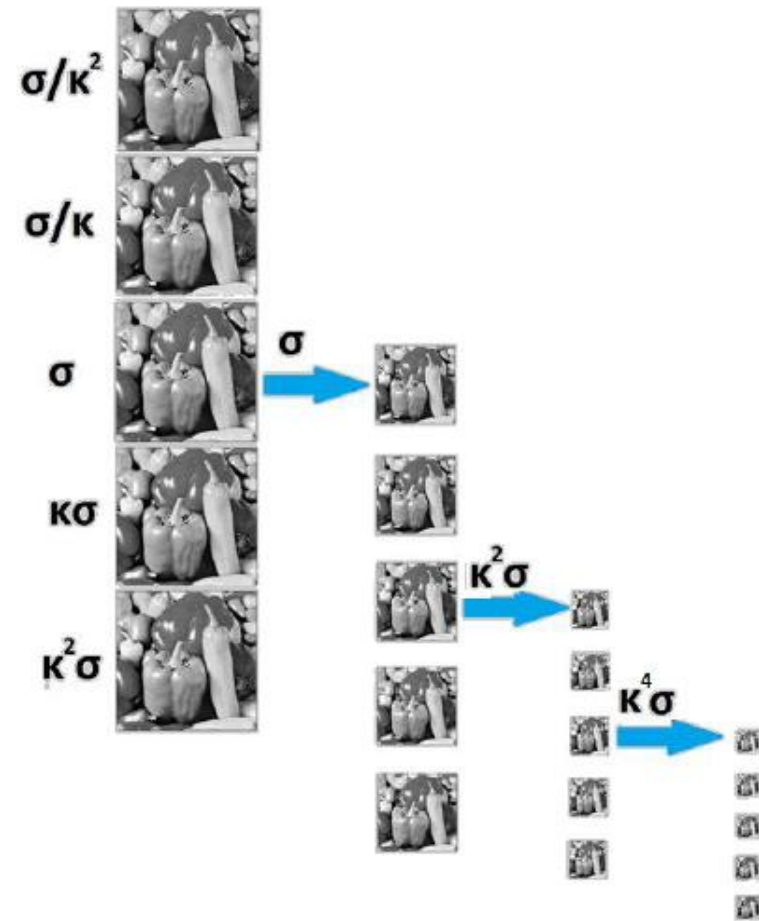
- So we can find corners.
- But how descriptive are these corners?
 - Not really,
 - Think that the roof of a building has corners
 - And your desk has corners...
- Finally, a revelation:

Lindeberg, T. 1994. Scale-space theory: A basic tool for analysing structures at different scales. *Journal of Applied Statistics*, 21(2):224-270

Lindeberg, Tony (1998). "Feature detection with automatic scale selection". *International Journal of Computer Vision* 30 (2): 79-116.

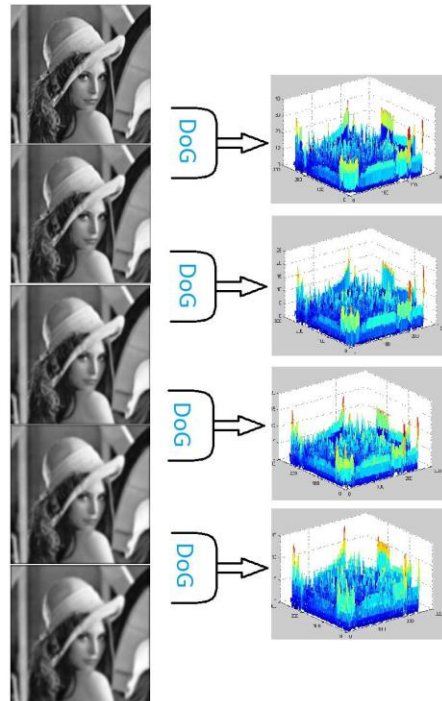
Feature Detection: Scale Space Theory

- We should be able to find these points which are prominent in different scales
- Create octaves with different scale among them
- Different blur level (Gaussian) in them



Feature Detection: Difference of Gaussians

- The main point is the difference of Gaussians
- We can then do this on our scale-space:



The mother of Features - SIFT

- Ok, we've seen how we can find interest points, but how about matching??
- David Lowe published the most influential paper in computer vision:

Lowe, D. Distinctive Image Features from Scale-Invariant Keypoints. International Journal of Computer Vision, 60, 2, pp 91-110 (2004).

- How many people do you think have cited this?
- >76000!!!!
(+ >25000 the conf paper)



Scale Invariant Feature Transform - SIFT

Contains all:

- Detection
- Description
- Matching

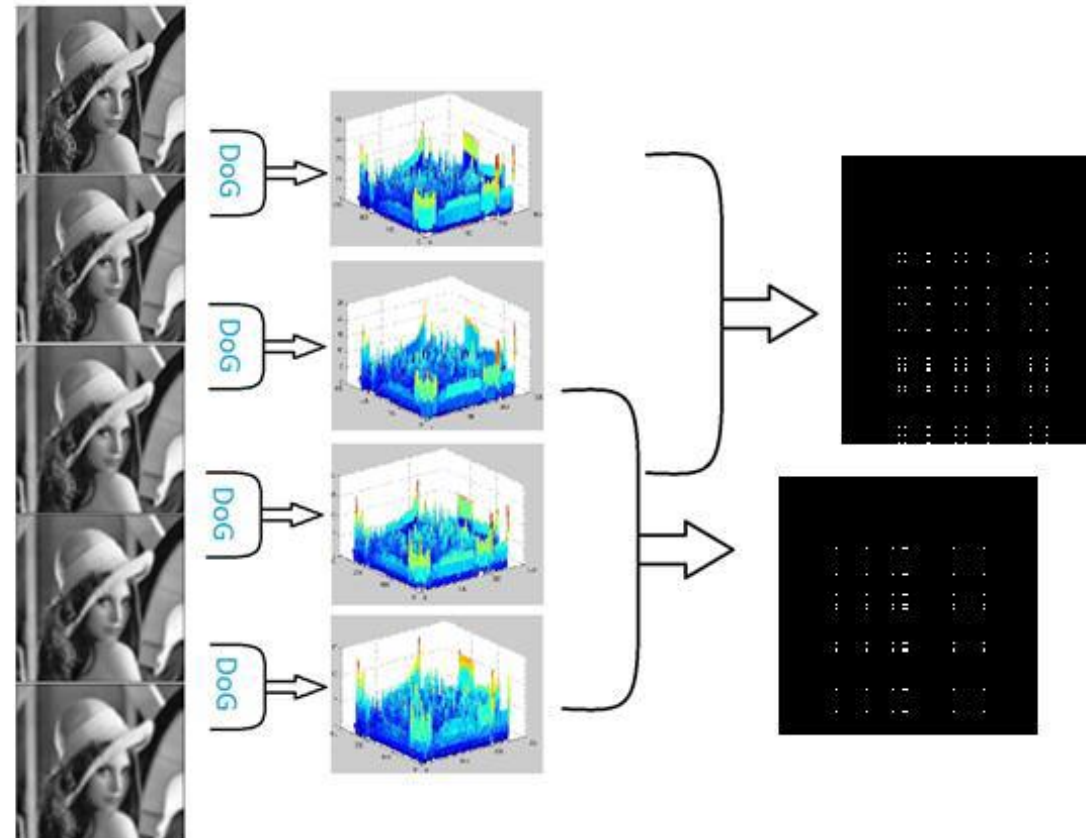
Main Feature is that it is robust in:

- Change of Translation
- Change in Scale
- Change in Rotation
- Change in 3D View Point
- Change in Illumination

Scale Invariant Feature Transform - SIFT

Detection

- Use DoG on Scale Space
 - Only the maximum or minimum in a neighborhood are considered
 - All octaves are investigated
- Specifically,
 - In groups of 3
 - 2 Set of Points from each octave
 - 4 octaves -> 8 set of Points



Scale Invariant Feature Transform - SIFT

Description

- Orientation:

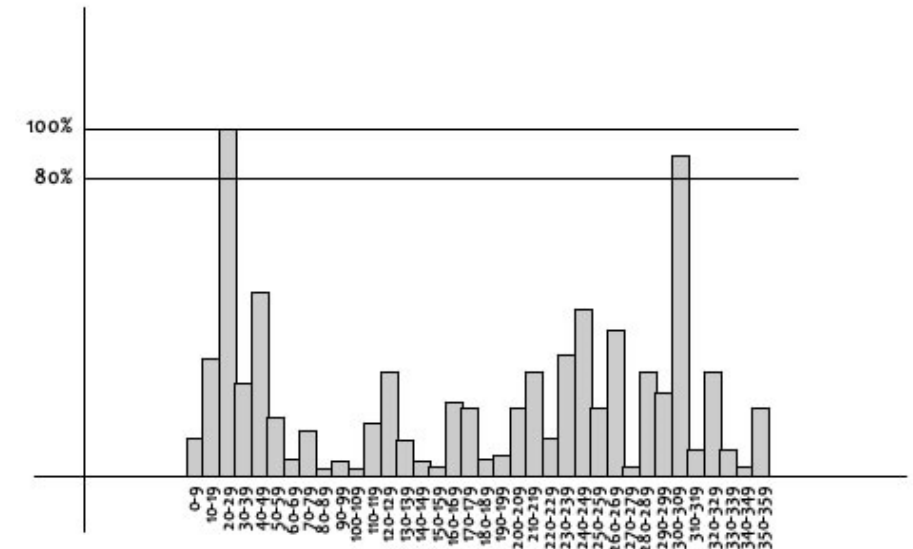
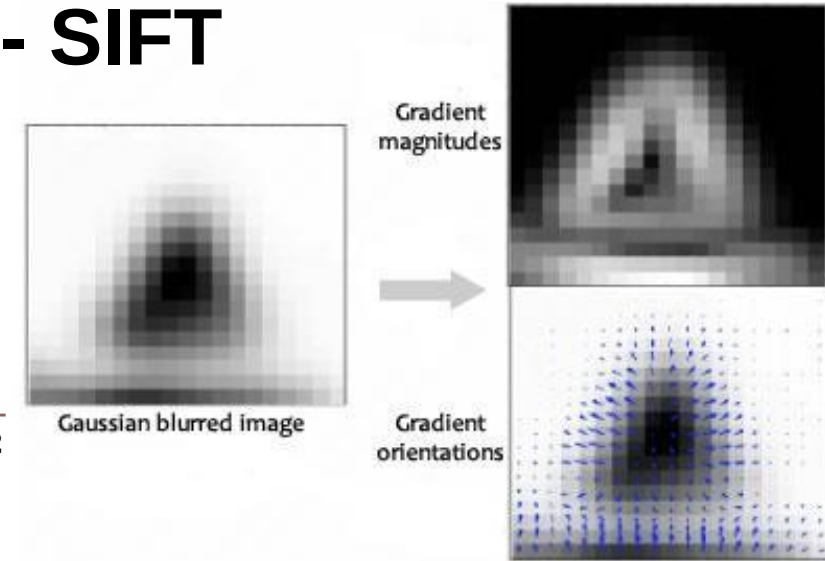
- Based on the gradient

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2}$$

$$\theta(x, y) = \tan^{-1} \left(\frac{L(x, y+1) - L(x, y-1)}{L(x+1, y) - L(x-1, y)} \right)$$

- Create a histogram of orientations
Consisting of 36 bins (every 10 degrees)

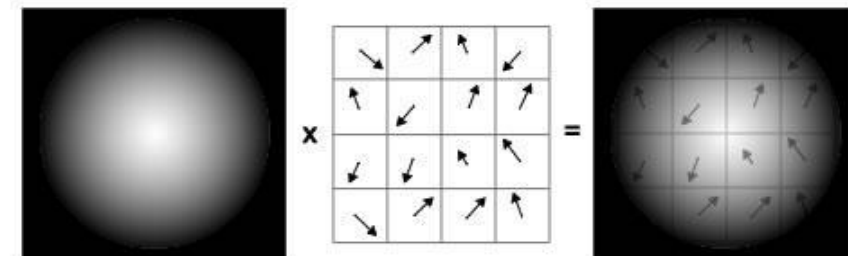
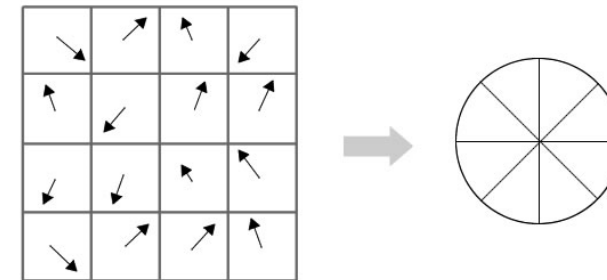
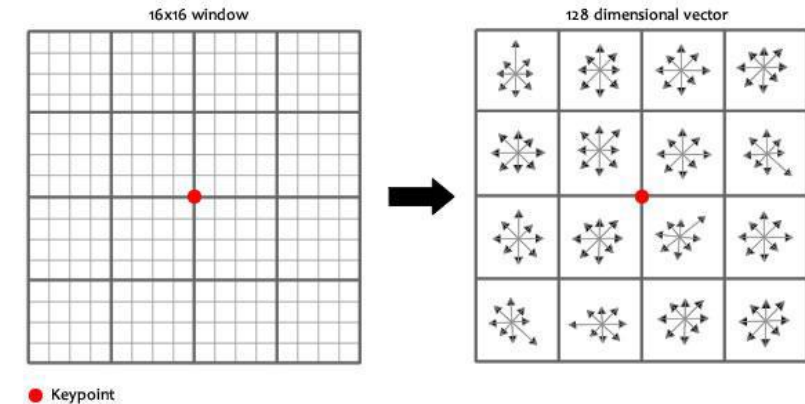
- Fingerprint



Scale Invariant Feature Transform - SIFT

Description

- Orientation
- Fingerprint:
 - Assume a 16 x 16 area around each Key Point
 - Create histogram with 8 bins (as before)
 - This gives out 128 values
 - Note that the values are scaled for proximity



Scale Invariant Feature Transform - SIFT

Matching

- Given some features from (let's say) 2 Images:
- Lowe proposed the ALL-ALL Euclidean distance:

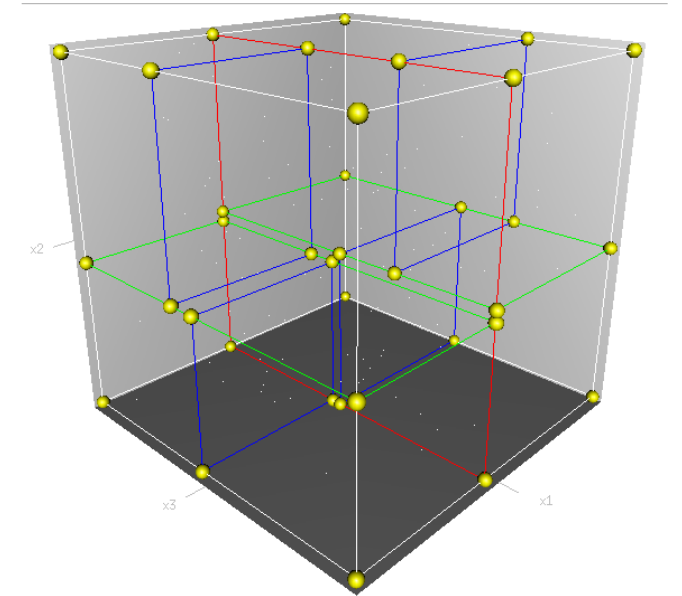
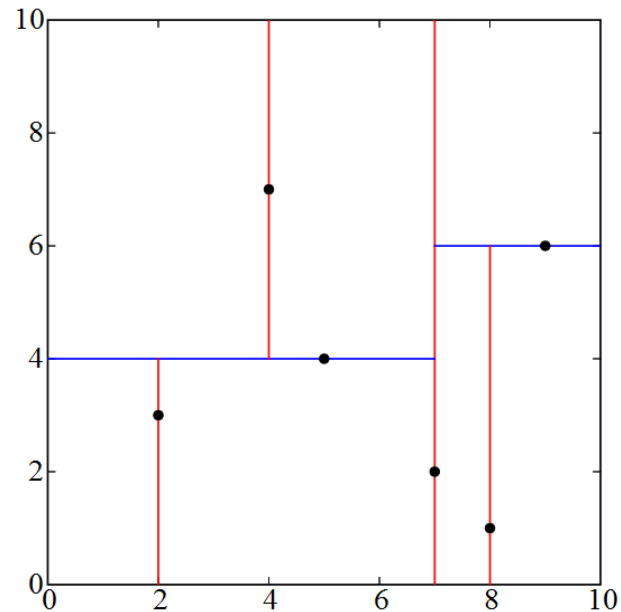
$$d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + (p_3 - q_3)^2 + \dots + (p_i - q_i)^2 + \dots + (p_n - q_n)^2}$$

- For a 640x640 image we can get even 1000 features
- With 128 values each feature, you see how this can get ugly quickly

Scale Invariant Feature Transform - SIFT

Matching

- So, he proposed the usage of Kd-Trees:
- Creation:
 - You split the space in half based on distance
 - Then again
 - Then again,....
- Search:
 - Start from top
 - Is it closer to 1 than 2?
 - Go in 1
 - Is it closer to 1.1 than 1.2?
 - Go in 1.1
 - GO On until you find a feature



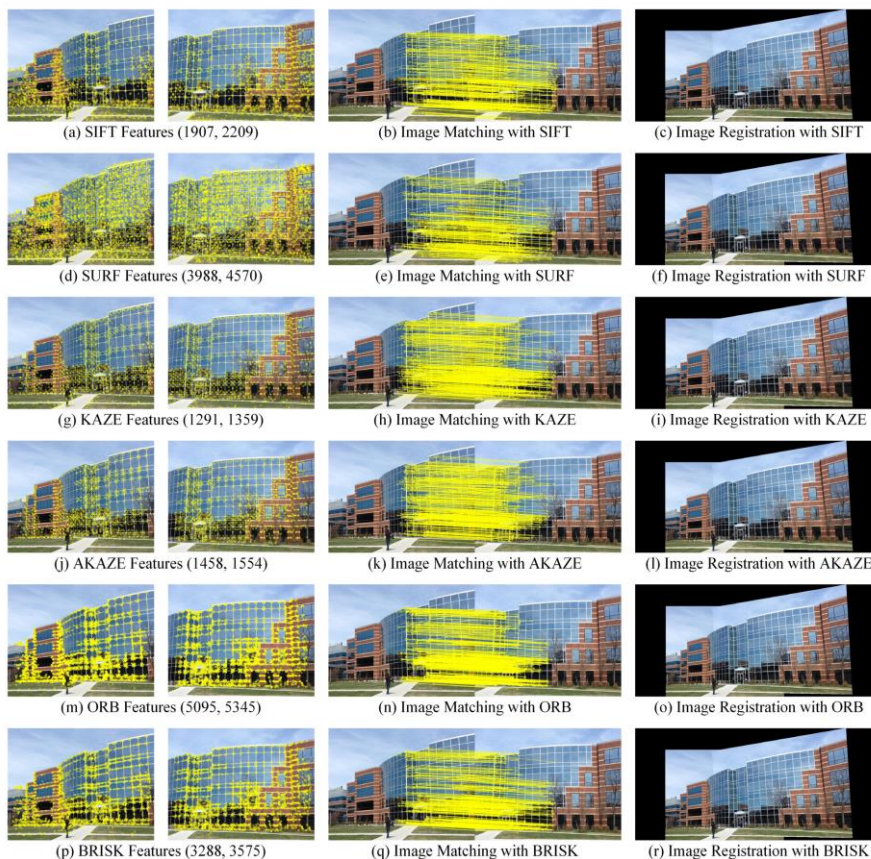
Feature points and areas

- A multitude of features have been proposed (specific for each application):

- SIFT
- SURF
- BRISK
- FREAK
- MSER
- ORB
- ...

- Patches

A special case to research



What applications do the Features have?

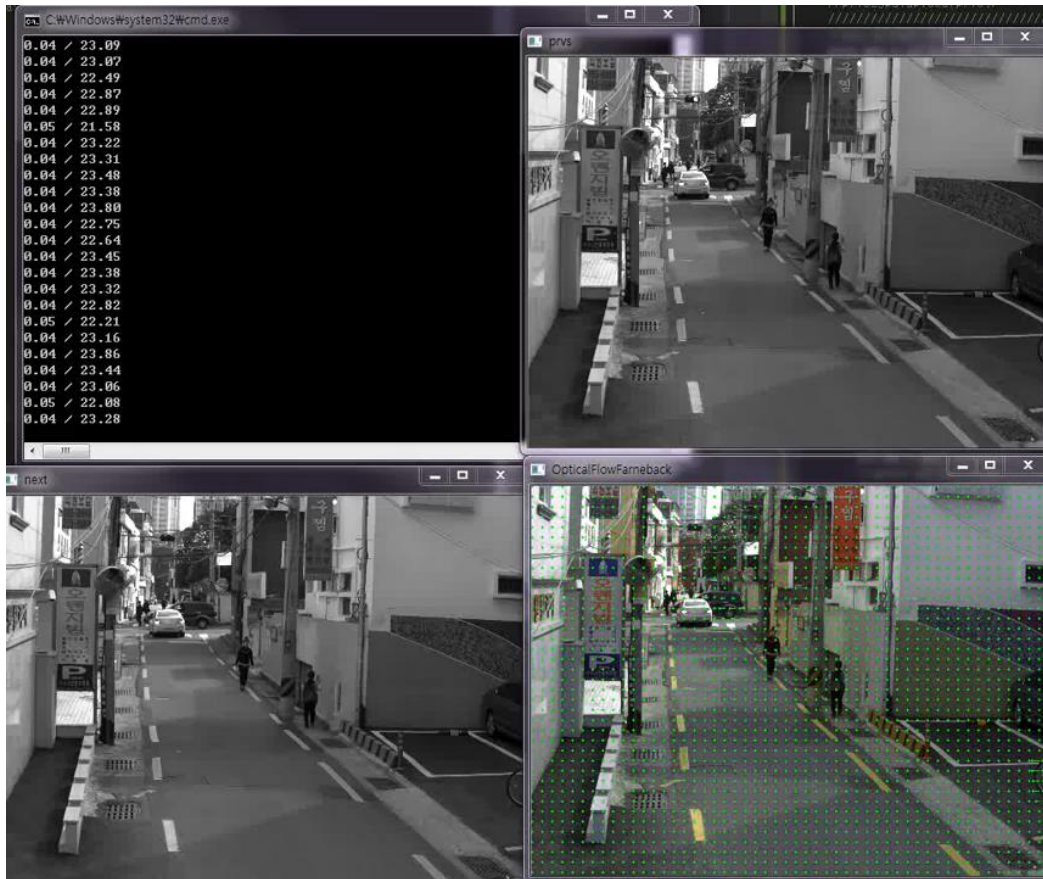
- More or less everything..
- Motion Estimation
- Localization
- Mapping
- Photogrammetry
- Image Retrieval
- Machine Learning
 - Object Detection
 - & Recognition
- Autonomous Driving
-
- More or less everything!



Image Feature Detection and Description

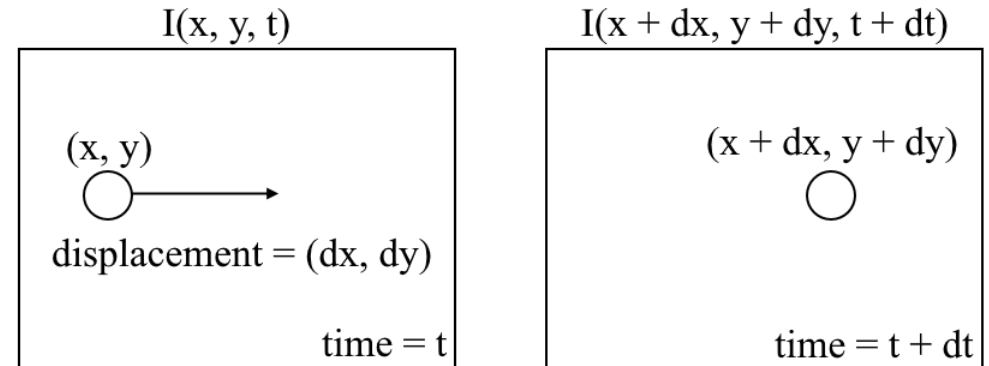
- What did we learn?
 - Is that a good feature?
 - How to get scale and rotational invariance in the features we get?
 - How to detect points of interest?
 - How to describe the feature of points so,
 - We can match them across multiple images.
 - How can we use features?

One slide on Optical Flow



One slide on Optical Flow

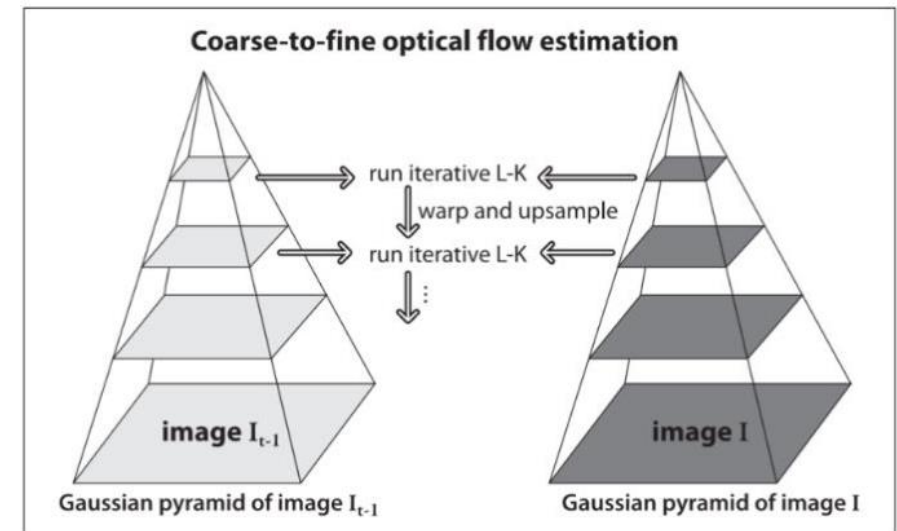
- Optical Flow is the apparent motion of objects due to the relative motion of the observer/object:



$$I(x, y, t) = I(x + \delta x, y + \delta y, t + \delta t)$$

- Sparse Optical Flow (original Lukas-Kanade)
 - Assume that the possible displacement is small and bounded
 - Assume that the displacement is homogeneous (constant inside the window)
 - Form normal equations for all pixels and solve using linear least squares.

- Dense Optical Flow (Gunnar-Farneback) :
 - is the implementation of Lukas Kanade on a pyramid scheme



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